

Title: Navigating the pitfalls of applying machine learning in genomics

Aim

The primary aim of the document is to identify and illustrate common pitfalls encountered when applying supervised machine learning (ML) to genomic data and to provide practical solutions for mitigating them.

The authors focus on the following key objectives:

- **Highlighting Violations of Assumptions:** The review seeks to show where the structure of biological data frequently violates the standard statistical assumptions of ML software, such as the assumption that examples are independent and identically distributed (IID).
- **Exploring Data Bias:** It explores how the inherent complexities of genomics data can bias performance evaluations and predictions, leading to inflated accuracy reports that do not reflect true generalisation.
- **Providing Mitigation Strategies:** For each identified pitfall, the document describes specific solutions and appropriate use cases where ML modelling shows significant scientific potential.
- **Quantitative Demonstration:** The researchers aim to provide a quantitative understanding of these issues through interactive notebooks containing data and code, allowing users to observe the effects of these pitfalls and test mitigation techniques in real-time.

Structurally, the document is organised around five specific pitfalls: distributional differences, dependent examples, confounding, leaky preprocessing, and unbalanced classes. Ultimately, the goal is to increase the trustworthiness and usefulness of machine learning within the field of biomedical research.

Theoretical Framework / Key Concepts

The theoretical framework for applying machine learning (ML) in genomics centers on supervised learning, where the primary objective is to develop a model that makes accurate predictions on new data or provides mechanistic insights by identifying predictive genomic features.

Core Machine Learning Definitions

- **Examples:** Also referred to as "samples" or "observations," these are the basic units of measurement and the primary data objects manipulated by an ML system.
- **Features (Covariates):** These are the properties of a given example, such as sequence patterns or gene expression values, used to make predictions.
- **Outcome (Label):** This represents what the analyst wants to predict, which can be a categorical label (classification) or a real number (regression).
- **Data Splits:** The process involves a training set used to fit the model, a disjoint test set used to evaluate performance, and a prediction set (or prospective validation set) where the outcomes are truly unknown.
- **Generalization Error:** This is a measure of how accurately a model predicts outcomes in data it has never seen before.

Fundamental Statistical Assumptions

The mathematics of common ML models and evaluation techniques like cross-validation rely on the assumption that data are independent and identically distributed (IID).

- **Independent:** The value of one example does not depend on or correlate with the values of others.
- **Identically Distributed:** All examples are generated by the same underlying distribution, sharing the same mean, variance, and shape.

Key Concepts and Pitfalls

The sources highlight that biological systems frequently violate these assumptions, leading to several specific pitfalls:

- **Distributional Differences:** This occurs when examples are not identically distributed, meaning the probability of observing a value varies across examples - often due to batch effects, inherent biological structure, or differences in study design.
- **Dependent Examples:** Biological data is replete with dependence, such as adjacent genomic positions with correlated activity or proteins within the same family or complex that share similar functions.
- **Confounding:** A confounder is an unmeasured or artefactual variable that induces dependence between features and the outcome, creating spurious associations that the model may mistakenly learn as direct biological effects.
- **Information Leakage ("Double Dipping"):** This is a subtle problem where information from the test set is inadvertently used during the processing of the training set, often during leaky preprocessing steps like global standardization or feature selection performed outside of cross-validation.
- **Unbalanced Classes:** This occurs when examples are not evenly distributed across outcome values, which can lead models to over-learn the majority class and fail to accurately predict the minority class of interest.

Methodological Concepts

The methodological concepts in the sources focus on how to properly evaluate machine learning (ML) models and the specific techniques used to mitigate various statistical pitfalls in genomics.

Performance Evaluation and Visualization

The sources define several standard procedures and metrics used to quantify how well a model generalizes to new data:

- **Cross-Validation:** A procedure where data is randomly split into k groups (folds); the model is fit on $k-1$ folds (the training set) and evaluated on the remaining group (the test set).
- **Group k -fold Cross-Validation (Blocking):** A variant of cross-validation that accounts for dependent examples (e.g., genes on the same chromosome or samples from the same patient) by ensuring all related examples stay within the same fold to prevent information leakage.
- **Performance Statistics:** Binary classifiers are summarized using metrics like accuracy, precision, recall (true positive rate), false positive rate (FPR), and the false discovery rate (FDR).
- **Performance Curves:**
 - **ROC Curve:** Plots the true positive rate against the FPR. The auROC quantifies performance, where 1.0 is perfect and 0.5 is random chance.
 - **Precision-Recall (PR) Curve:** Plots precision against recall. The auPR is often preferred for unbalanced problems where the minority class is of primary interest.

Mitigation and Correction Strategies

The sources outline specific methodological approaches to address the five identified pitfalls:

- **Handling Distributional Differences:** Methods include batch correction (e.g., quantile normalization, ComBat), surrogate variable analysis, and adversarial learning to make models robust to different biological or technical contexts.
- **Addressing Dependent Examples:** Beyond blocking, researchers may use mixed effects models, time-series models, or autocorrelation models to explicitly model the covariance between examples.
- **Reducing Confounding:** Strategies include randomizing examples across potential confounders, using principal component analysis (PCA) or PEER to capture unmeasured structure, and explicitly including confounding variables in the ML model.

- Preventing Leaky Preprocessing: The primary methodology is to perform all data transformations - such as scaling, imputation, or feature selection - strictly within the training fold of cross-validation. Parameters (like mean and standard deviation) must be learned from the training set and then applied to the test set independently.
- Managing Unbalanced Classes: Three common strategies are used to emphasize the minority class:
 - Oversampling: Duplicating existing minority data or synthesizing "plausible" new examples (e.g., SMOTE).
 - Undersampling: Taking a random subset of the majority class to match the minority size.
 - Weighting: Using the inverse of class proportions to give more weight to errors in the minority class.

Statistical Detection Methods

To check if model assumptions are being met, the sources recommend:

- Visualization: Using scatter plots or histograms to compare feature distributions across different datasets.
- Statistical Tests: Applying tests like the binomial test (for binary features), the Kolmogorov–Smirnov test (univariate), or Maximum Mean Discrepancy (multivariate) to detect dataset shift.
- Graph Visualization: Rendering tabular data as a graph to identify hub nodes and dependencies before applying ML tools.

Assumptions

The primary statistical assumption underlying supervised machine learning (ML) models and their evaluation techniques is that data examples are independent and identically distributed (IID).

According to the sources, this foundational assumption can be broken down into two components:

- Independent: This assumes that the value of one example does not depend on or correlate with the values of other examples.
- Identically Distributed: This assumes that every example is generated by the same underlying probability distribution, meaning they share a particular mean, variance and shape.

Why These Assumptions Matter

These assumptions are critical because the mathematics used in common ML toolkits and performance evaluation procedures, such as cross-validation, depend on them to provide accurate results. If these assumptions are inaccurate for a specific dataset, the reported predictive accuracy may be substantially inflated compared to how the model would actually perform on a completely independent prediction set.

Common Violations in Genomics

The sources highlight that biological systems frequently violate these assumptions in several ways:

- Violations of Independence: Biological data often contains dependencies, such as adjacent genomic positions with correlated activity or proteins within the same family, pathway, or complex that have very similar functions.
- Violations of Identical Distribution: Examples may not be identically distributed due to inherent biological structure (e.g., differences between euchromatin and heterochromatin), population genetic structure (ancestry), or technical factors like batch effects.

Importantly, the sources note that ML software cannot detect these violations; it is the responsibility of the modeller to verify these assumptions and adjust their statistical procedures accordingly.

Methodology

The methodology employed in the source is a comprehensive review that combines theoretical analysis, illustrative case studies, and quantitative demonstrations to identify and mitigate common errors in genomic machine learning.,,

The authors structured their approach around the following key components:

Framework of Pitfall Analysis

The Review is organised around five specific pitfalls: distributional differences, dependent examples, confounding, leaky preprocessing, and unbalanced classes. For each problem, the authors follow a consistent methodological pattern:

- General Description: Defining the pitfall and the statistical assumptions it violates.,
- Illustrative Examples: Providing real-world scenarios from the literature where these pitfalls frequently occur, such as in GWAS, single-cell experiments, or proteomics.,
- Quantitative Demonstration: Using data and code to measure the actual impact of these errors on model performance.
- Mitigation Strategies: Offering specific statistical or procedural solutions, such as batch correction, blocking, or adversarial learning.,

Quantitative Demonstrations via Interactive Notebooks

A central part of the methodology involved the development of interactive computational notebooks., These tools allow researchers to:

- Use Synthetic and Real Data: The notebooks utilize either simulated data - designed with no true associations to test for false positives - or real epigenomic data.,
- Simulate Bias: For instance, the authors used simulations of random bipartite graphs to show how dependent examples falsely inflate performance metrics like the area under the precision–recall (auPR) curve.
- Test Mitigation: They demonstrate how techniques like group k-fold cross-validation (blocking) can bring reported performance back to realistic levels.,

Case Study Analysis

The authors performed in-depth analyses of specific genomic tasks to illustrate their points:

- Transcription Factor Binding: They examined how models trained on *in vitro* data often fail to generalise to *in vivo* contexts due to distributional differences.
- 3D Chromatin Interactions: They reviewed how genomic distance can act as a confounder in predicting Hi-C data from epigenetic features.
- DNA Methylation: They highlighted "leaky preprocessing" in methylation studies where feature selection is often performed on the entire dataset before cross-validation.,

Evaluation of Model Performance

The methodology includes a rigorous definition and evaluation of standard ML procedures:

- Cross-Validation Mechanics: Detailing the random splitting of data into folds to estimate performance on unseen data.
- Performance Metrics: Comparing the utility of various statistics, such as accuracy, precision, recall, and auROC, specifically noting that auPR is generally preferred for the unbalanced problems common in genomics.,

- The "Remedy" Approach: The authors conclude by proposing a methodological standard: researchers must ensure the relationship between training and test sets is identical to the relationship between training and the final prediction set.

Main Findings

The main findings of the sources highlight that while machine learning (ML) has great potential in genomics, its application is frequently undermined by violations of fundamental statistical assumptions, specifically the requirement that data be independent and identically distributed (IID). These violations lead to several critical issues:

1. Performance Inflation

A central finding is that when modeling assumptions are inaccurate, the reported predictive accuracy of a model is often substantially inflated compared to its true generalization error on a completely independent prediction set.

- Dependent Examples: Failing to account for biological dependencies (like shared protein families or genomic proximity) allows information to "leak" across the train–test divide, which can falsely elevate the area under the precision–recall (auPR) curve by more than 0.5.
- Leaky Preprocessing: Performing data transformations (like scaling or feature selection) on an entire dataset before splitting it into training and test folds leads to "double dipping." This can produce striking increases in model accuracy even on synthetic datasets with no real biological signal.

2. Generalization Failures across Contexts

The sources find that distributional differences (Pitfall 1) between training and application settings cause significant performance drops.

- Case Study Evidence: Models learned from *in vitro* data often perform much better in cross-validation than on *in vivo* data, with auROC differences of up to 0.4.
- Technical Biases: Differences in sequencing depth, reagents, or personnel (batch effects) shift data distributions, meaning a model trained on a mixture of batches will fail when applied to a new, unseen batch.

3. Misinterpretation due to Confounding

The researchers find that unmeasured or artefactual variables (confounders) create associations that ML models mistakenly learn as direct biological effects.

- Biological Risks: In genetic studies, unmodelled population structure (ancestry) can lead to irreproducible results, where a model predicts a disease like autism based on shared ancestry rather than actual genetic markers for the condition.
- Technical Risks: Variables like sequencing depth act as nonlinear confounders; if not adjusted for, models will systematically mispredict signal values when applied to data with different depths.

4. Necessity of Specific Evaluation Metrics

The sources demonstrate that standard metrics like accuracy and auROC can be highly misleading in genomics due to unbalanced classes.

- In highly imbalanced problems - such as predicting enhancers that constitute only 1% of genomic windows - high accuracy can be achieved by simply always predicting the negative class.
- The authors find that auPR is generally preferred for these problems because it measures how recall increases as the false discovery proportion increases, providing a more realistic assessment of performance when false positives dominate.

5. The "Unified Remedy"

Ultimately, the document finds that the solution to these interrelated pitfalls is to construct training, test, and prediction sets such that the relationship between training and testing mirrors the relationship between training and the final real-world application. For example, if the final prediction set is a new batch of data, the test set used for evaluation should also be composed of a separate batch.

Limitations

The limitations of applying machine learning (ML) to genomics stem from the fact that biological systems frequently violate the fundamental statistical assumptions upon which common ML toolkits and evaluation procedures are built.

Violation of Statistical Assumptions

The most critical limitation is the assumption that data are independent and identically distributed (IID).

- Independence Violations: Biological data is "replete with violations" of independence. For example, genomic proximity causes correlated activity in adjacent positions, and protein families share similar functions, creating inherent dependencies that ML software cannot automatically detect.
- Identical Distribution Violations: Data is often not identically distributed due to batch effects (caused by differences in reagents, personnel, or instruments), population genetic structure (ancestry), or inherent biological structure like the differences between euchromatin and heterochromatin.

Performance Evaluation Pitfalls

Methodological errors often lead to substantially inflated predictive accuracy that does not reflect true generalization.

- Leaky Preprocessing: A pervasive limitation is "double dipping," where information from the test set leaks into the training set during global transformations like scaling, imputation, or feature selection. This can produce high accuracy even on random data with no real biological signal.
- Unbalanced Classes: In genomics, imbalance is the "default state" (e.g., enhancers may constitute only 1% of the data). This can lead models to over-learn the majority class, making standard metrics like accuracy misleading - for instance, 99% accuracy can be achieved simply by always predicting the negative class.
- Inflation via Dependence: Failing to account for dependencies (e.g., hub nodes in interaction networks) can falsely elevate performance metrics like the area under the precision-recall (auPR) curve by more than 0.5.

Data and Interpretation Limitations

- Confounding Variables: Unmeasured or technical variables (like sequencing depth) can create spurious associations. ML models may estimate an association that modellers wrongly interpret as a direct biological effect, leading to failures when the model is applied in a new context.
- Limited Scientific Value: Some ML formulations have limited utility, such as circular problems where the label being predicted is already encoded in the features, or tasks that predict cheap data from expensive data that is already available.
- Distributional Shifts: Models learned in one context (e.g., *in vitro* assays) often fail to generalize to another (e.g., *in vivo* contexts), with performance drops in auROC of up to 0.4.

Software and Oversight Constraints

- Software Inability to Detect Pitfalls: Current ML software is unable to detect when modelling assumptions are inaccurate; the responsibility rests entirely on the modeller to adjust statistical procedures.
- Need for Scepticism: The trustworthiness of ML in biomedical research is limited if results are accepted at face value without healthy scepticism and robust follow-up analyses.

Implication of Practise

The provided sources outline a comprehensive framework for robust genomic machine learning, emphasizing that practitioners must move beyond standard automated workflows to ensure models remain reliable in real-world forecasting contexts. The framework focuses on rigorous validation, technical adjustments, and continuous monitoring.

Pre-Modelling Audits and Data Assumptions

Before any modelling begins, practitioners are responsible for manually verifying Independence and Identical Distribution (IID) assumptions. Since software cannot detect these violations, visual inspections (histograms, scatter plots) and statistical tests, such as the Kolmogorov-Smirnov test, should be used to identify distributional differences across batches or sites. Failure to perform this audit is cited as a primary cause of inflated performance claims in genomics.

Rigorous Validation and Evaluation

Standard validation techniques often produce misleadingly optimistic results. To combat this, the sources recommend:

- **Matching Train-Test to Deployment:** The relationship between training and test sets must mirror the real-world forecasting context. If a model will be applied to a new hospital site or sequencing batch, the test set must come from a similarly distinct source rather than a random split.
- **Group k-Fold Cross-Validation:** When biological dependencies exist - such as shared patient data or related protein families - group k-fold cross-validation should be the default to prevent performance inflation of over 0.5 in auPR.
- **Strict Preprocessing Pipelines:** All data transformations (scaling, imputation, feature selection) must be fitted exclusively on the training fold and applied to the test fold to avoid "leaky preprocessing".

Technical Adjustments and Confounder Management

To ensure models learn genuine biological signals rather than technical artefacts, practitioners must:

- **Identify and Adjust for Confounders:** Variables like sequencing depth, batch identity, and population ancestry can create spurious associations. These should be included as model inputs or adjusted via PCA/PEER factors to ensure results are reproducible.
- **Apply Batch Correction:** Before pooling data, methods like quantile normalisation or ComBat should be used. A model is only considered robust if a classifier cannot distinguish between different batches at a level above chance.
- **Address Class Imbalance:** Since genomic data is often imbalanced (e.g., rare disease variants), practitioners should use oversampling (SMOTE), undersampling, or class frequency weighting to ensure the model does not fail on the minority class.

Performance Metrics and Interpretation

The sources advocate for a shift in how model success is measured:

- **Prioritise auPR over auROC:** In imbalanced clinical contexts, Area Under the Precision-Recall curve (auPR) is a more informative metric for operational decisions than auROC.
- **Realistic Performance Expectations:** Performance on *in vitro* assay data should be viewed as an optimistic upper bound rather than a deployment estimate; *in vivo* performance can be up to 0.4 auROC lower.

Long-term Reliability and Scepticism

Model development is treated as an ongoing cycle rather than a one-time event. Practitioners should build interactive validation notebooks to continuously monitor for distributional drift as technology and patient populations evolve. Finally, a "systematic scepticism" should be applied to all published claims, evaluating them against a checklist of these robust principles before adoption.